EcoVision AI

AI - Driven
Forest Monitoring and Analytics

By Shooting Stars

HACKaMINeD 2026

HACKaMIND 2026

Track: AI - Driven Forest Monitoring and Analytics

Track Sponsor: Amnex

AMNEX

BINGHAMTON UNIVERSITY
STATE UNIVERSITY OF NEW YORK

Team Shooting Stars

- Yashvi Dalsaniya
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The Challenge

Forests are essential to planetary health, serving as climate regulators, biodiversity reservoirs, and carbon sinks. Yet comprehensive forest monitoring remains a persistent challenge for environmental scientists and forestry managers worldwide

AIM:Converting raw aerial imagery into actionable ecological intelligence at scale.

Manual Monitoring:

Requires significant manpower

Is prone to errors

Is slow and expensive

Requires expertise in domain

Significant Problems:

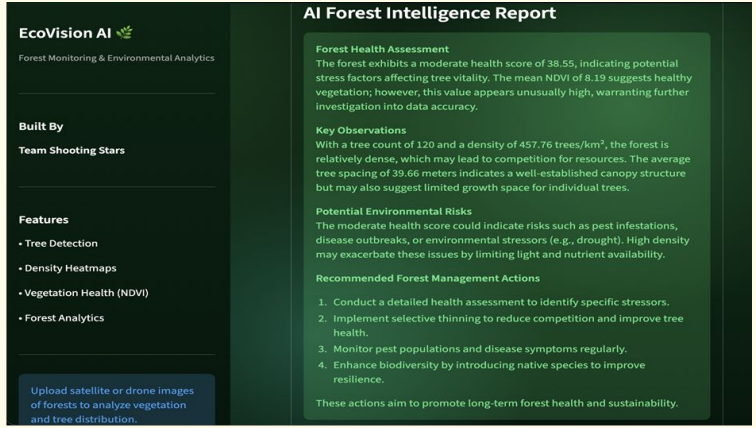
No/Low Availability of Data

Lack of Adaptive Solution

Lack of human-understandable output

Proposed Solution

EcoVision AI transforms raw aerial imagery into comprehensive forest analytics and management insights. Our AI-powered system automatically detects individual trees, measures forest density, analyses vegetation health, clusters tree species, and generates detailed ecological reports.



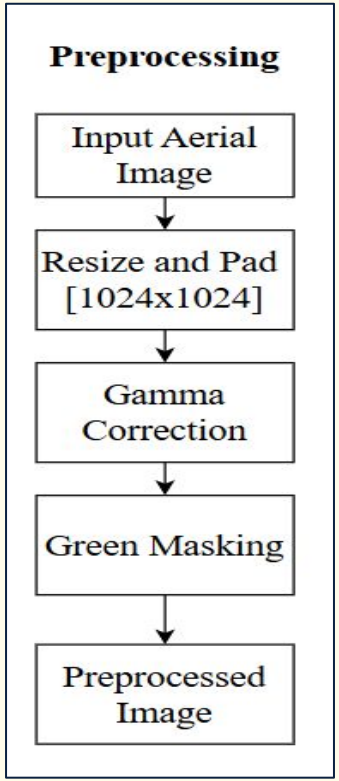
Tree Detection
Identifies individual tree crowns from aerial imagery

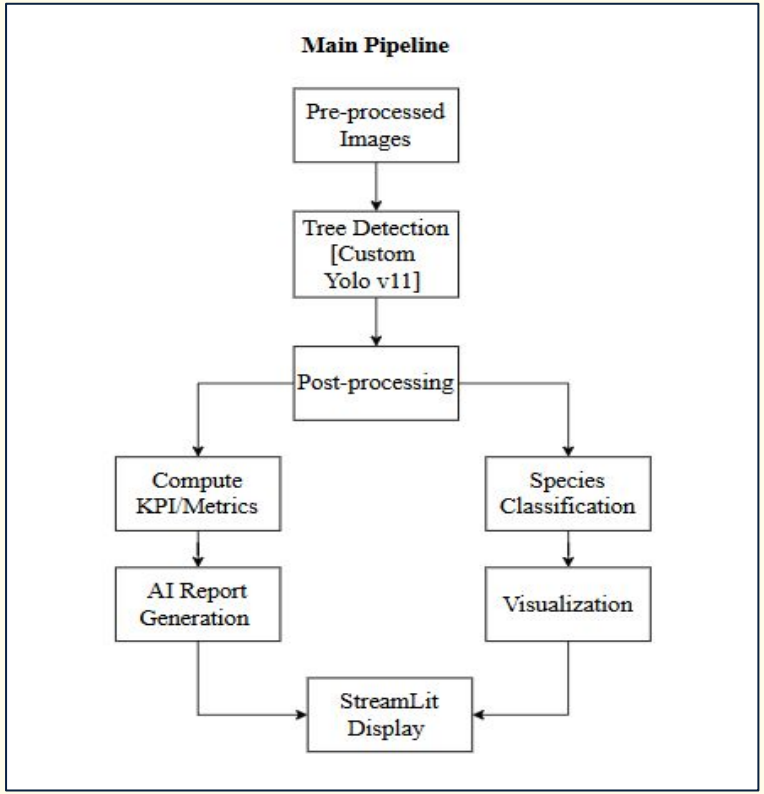
Health Assessment
Evaluates vegetation condition using NDVI indices

Density Analysis
Calculates spatial distribution patterns

Species Clustering
Groups trees into biodiversity categories

Architecture





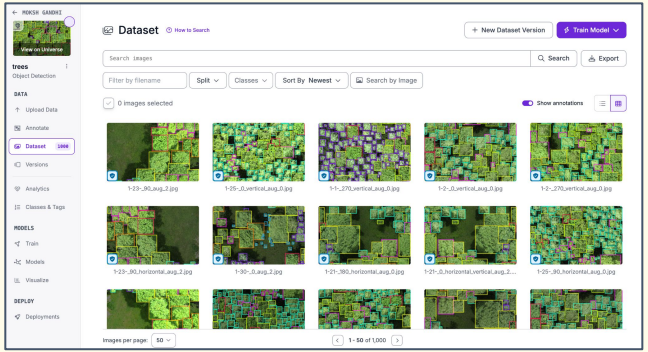
Technology Stack:

- Streamlit
- OpenCv
- PyTorch
- Ultralytics YOLO
- Scikit-Learn
- NumPy
- Pandas
- SciPy
- Matplotlib
- Pillow (PIL)
- OpenAI API

Model Logic

Model Logic

1. Dataset Annotation using Roboflow
2. Data Preprocessing and Augmentation
3. YOLOv11 Model Training
4. Tree Detection using Trained Model
5. Tree Counting and Output Integration



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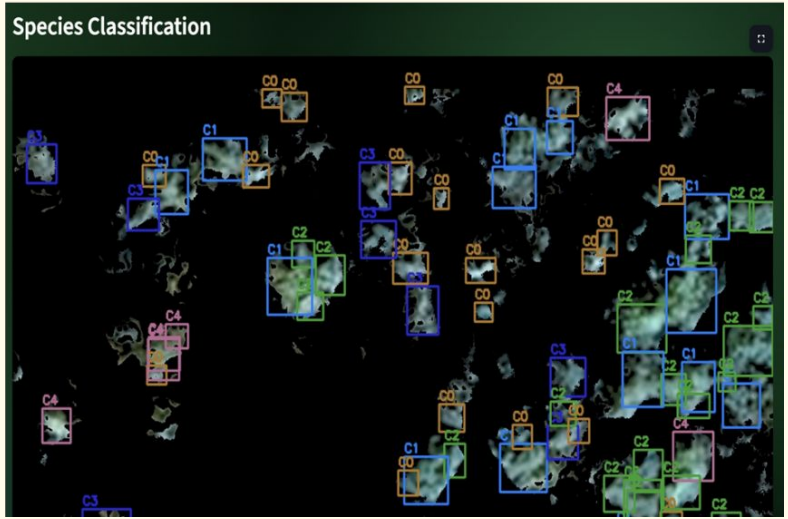

Impact and Future Vision

Impact

- Enables **automated large-scale forest monitoring** using AI.
- Helps detect **deforestation, vegetation stress, and forest degradation early**. Supports **data-driven decisions** for conservation and forest management.
- Can assist governments and NGOs in **climate and sustainability initiatives**.

Future Vision

- Integrate **satellite multispectral data for accurate NDVI analysis**.
- Build a **real-time monitoring system using drones and satellite feeds**.
- Improve **AI models for higher accuracy across different forest types**.
- Integrate with **GIS platforms for interactive environmental dashboards**.

Impact and Future Vision

Thank You!

EcoVision AI
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Questions?

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